В этом задании Вам предстоит обучить языковую модель с помощью рекуррентной нейронной сети. В отличие от семинарского занятия, Вам необходимо будет работать с отдельными словами, а не буквами.

Установим модуль `datasets`, чтобы нам проще было работать с данными.

```
# !pip install datasets
```

Импорт необходимых библиотек

```
import random
import numpy as np
import torch
import os

# Фиксация всех источников случайности
def set_seed(seed=42):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)

    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False

    os.environ['PYTHONHASHSEED'] = str(seed)

set_seed(42)
```

```python
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader

import matplotlib.pyplot as plt

from tqdm.auto import tqdm
from datasets import load_dataset
from nltk.tokenize import sent_tokenize, word_tokenize
from sklearn.model_selection import train_test_split
import nltk

from collections import Counter
from typing import List

import math

import seaborn
seaborn.set_theme(palette='summer')

nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to C:\Users\pong/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
```

```
True
```

```python
device = 'cuda' if torch.cuda.is_available() else 'cpu'
device
```

```
'cuda'
```

Подготовка данных

Воспользуемся датасетом imdb. В нем хранятся отзывы о фильмах с сайта imdb. Загрузим данные с помощью функции `load_dataset`

```python
# Загрузим датасет
dataset = load_dataset('imdb')
```

Препроцессинг данных и создание словаря (1 балл)

Далее вам необходмо самостоятельно произвести препроцессинг данных и получить словарь или же просто `set` строк. Что необходимо сделать:

1. Разделить отдельные тренировочные примеры на отдельные предложения с помощью функции `sent_tokenize` из бибилиотеки `nltk`. Каждое отдельное предложение будет одним тренировочным примером.
2. Оставить только те предложения, в которых меньше `word_threshold` слов.

3. Посчитать частоту вхождения каждого слова в оставшихся предложениях. Для деления предлоения на отдельные слова удобно использовать функцию `word_tokenize`.
4. Создать объект `vocab` класса `set`, положить в него служебные токены '<unk>', '<bos>', '<eos>', '<pad>' и vocab_size самых частовстречающихся слов.

```
display(dataset['train']['text'][:7])
```

```
['I rented I AM CURIOUS-YELLOW from my video store because of all the
controversy that surrounded it when it was first released in 1967. I
also heard that at first it was seized by U.S. customs if it ever
tried to enter this country, therefore being a fan of films considered
"controversial" I really had to see this for myself.<br /><br />The
plot is centered around a young Swedish drama student named Lena who
wants to learn everything she can about life. In particular she wants
to focus her attentions to making some sort of documentary on what the
average Swede thought about certain political issues such as the
Vietnam War and race issues in the United States. In between asking
politicians and ordinary denizens of Stockholm about their opinions on
politics, she has sex with her drama teacher, classmates, and married
men.<br /><br />What kills me about I AM CURIOUS-YELLOW is that 40
years ago, this was considered pornographic. Really, the sex and
nudity scenes are few and far between, even then it\'s not shot like
some cheaply made porno. While my countrymen mind find it shocking, in
reality sex and nudity are a major staple in Swedish cinema. Even
Ingmar Bergman, arguably their answer to good old boy John Ford, had
sex scenes in his films.<br /><br />I do commend the filmmakers for
the fact that any sex shown in the film is shown for artistic purposes
rather than just to shock people and make money to be shown in
pornographic theaters in America. I AM CURIOUS-YELLOW is a good film
for anyone wanting to study the meat and potatoes (no pun intended) of
Swedish cinema. But really, this film doesn\'t have much of a plot.',
 '"I Am Curious: Yellow" is a risible and pretentious steaming pile.
It doesn\'t matter what one\'s political views are because this film
can hardly be taken seriously on any level. As for the claim that
frontal male nudity is an automatic NC-17, that isn\'t true. I\'ve
seen R-rated films with male nudity. Granted, they only offer some
fleeting views, but where are the R-rated films with gaping vulvas and
flapping labia? Nowhere, because they don\'t exist. The same goes for
those crappy cable shows: schlongs swinging in the breeze but not a
clitoris in sight. And those pretentious indie movies like The Brown
Bunny, in which we\'re treated to the site of Vincent Gallo\'s
throbbing johnson, but not a trace of pink visible on Chloe Sevigny.
Before crying (or implying) "double-standard" in matters of nudity,
the mentally obtuse should take into account one unavoidably obvious
anatomical difference between men and women: there are no genitals on
display when actresses appears nude, and the same cannot be said for a
man. In fact, you generally won\'t see female genitals in an American
film in anything short of porn or explicit erotica. This alleged
double-standard is less a double standard than an admittedly
```

depressing ability to come to terms culturally with the insides of women\'s bodies.',
 "If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story.

One might feel virtuous for sitting thru it because it touches on so many IMPORTANT issues but it does so without any discernable motive. The viewer comes away with no new perspectives (unless one comes up with one while one's mind wanders, as it will invariably do during this pointless film).

One might better spend one's time staring out a window at a tree growing.

",
 "This film was probably inspired by Godard's Masculin, féminin and I urge you to see that film instead.

The film has two strong elements and those are, (1) the realistic acting (2) the impressive, undeservedly good, photo. Apart from that, what strikes me most is the endless stream of silliness. Lena Nyman has to be most annoying actress in the world. She acts so stupid and with all the nudity in this film,...it's unattractive. Comparing to Godard's film, intellectuality has been replaced with stupidity. Without going too far on this subject, I would say that follows from the difference in ideals between the French and the Swedish society.

A movie of its time, and place. 2/10.",
 'Oh, brother...after hearing about this ridiculous film for umpteen years all I can think of is that old Peggy Lee song..

"Is that all there is??" ...I was just an early teen when this smoked fish hit the U.S. I was too young to get in the theater (although I did manage to sneak into "Goodbye Columbus"). Then a screening at a local film museum beckoned - Finally I could see this film, except now I was as old as my parents were when they schlepped to see it!!

The ONLY reason this film was not condemned to the anonymous sands of time was because of the obscenity case sparked by its U.S. release. MILLIONS of people flocked to this stinker, thinking they were going to see a sex film...Instead, they got lots of closeups of gnarly, repulsive Swedes, on-street interviews in bland shopping malls, asinie political pretension...and feeble who-cares simulated sex scenes with saggy, pale actors.

Cultural icon, holy grail, historic artifact..whatever this thing was, shred it, burn it, then stuff the ashes in a lead box!

Elite esthetes still scrape to find value in its boring pseudo revolutionary political spewings..But if it weren\'t for the censorship scandal, it would have been ignored, then forgotten.

Instead, the "I Am Blank, Blank" rhythymed title was repeated endlessly for years as a titilation for porno films (I am Curious, Lavender - for gay films, I Am Curious, Black - for blaxploitation films, etc..) and every ten years or so the thing rises from the dead, to be viewed by a new generation of suckers who want to see that "naughty sex film" that "revolutionized the film industry"...

Yeesh, avoid like the plague..Or if you MUST see it - rent the video and fast forward to the "dirty" parts, just to get it over with.

',
 "I would put this at the top of my list of films in the category of

```
unwatchable trash! There are films that are bad, but the worst kind
are the ones that are unwatchable but you are suppose to like them
because they are supposed to be good for you! The sex sequences, so
shocking in its day, couldn't even arouse a rabbit. The so called
controversial politics is strictly high school sophomore amateur night
Marxism. The film is self-consciously arty in the worst sense of the
term. The photography is in a harsh grainy black and white. Some
scenes are out of focus or taken from the wrong angle. Even the sound
is bad! And some people call this art?<br /><br />",
 "Whoever wrote the screenplay for this movie obviously never
consulted any books about Lucille Ball, especially her autobiography.
I've never seen so many mistakes in a biopic, ranging from her early
years in Celoron and Jamestown to her later years with Desi. I could
write a whole list of factual errors, but it would go on for pages. In
all, I believe that Lucille Ball is one of those inimitable people who
simply cannot be portrayed by anyone other than themselves. If I were
Lucie Arnaz and Desi, Jr., I would be irate at how many mistakes were
made in this film. The filmmakers tried hard, but the movie seems
awfully sloppy to me."]
```

```
sentences = []
word_threshold = 32

# Получить отдельные предложения и поместить их в sentences

for sentence in tqdm(dataset['train']['text']):
    sentences.extend(
        [x.lower() for x in sent_tokenize(sentence) if
len(word_tokenize(x.lower())) < word_threshold]
        )
```

{"model_id":"24f09fa5ffac4a07b5839ce5b0acca28","version_major":2,"version_minor":0}

```
print("Всего предложений:", len(sentences))
```

```
Всего предложений: 198801
```

Посчитаем для каждого слова его встречаемость.

```
words = Counter()

# Расчет встречаемости слов
for sentence in tqdm(sentences):
    for word in word_tokenize(sentence):
        words[word] += 1
```

{"model_id":"4d019ee20e8a4b5dbf0fb05c15d0fff1","version_major":2,"version_minor":0}

```
len(words)
```

```
68415
```

Добавим в словарь `vocab_size` самых встречающихся слов.

```
vocab = set()
vocab_size = 40000

# Наполнение словаря

vocab = set(['<unk>', '<bos>', '<eos>', '<pad>'])

# Топ слов (длиной vocab_size = 40000)
words_most_common = [word for word, _ in
words.most_common(vocab_size)]
vocab.update(words_most_common)

len(vocab)
```

```
40004
```

```
assert '<unk>' in vocab
assert '<bos>' in vocab
assert '<eos>' in vocab
assert '<pad>' in vocab
assert len(vocab) == vocab_size + 4

print("Всего слов в словаре:", len(vocab))
```

```
Всего слов в словаре: 40004
```

Подготовка датасета (1 балл)

Далее, как и в семинарском занятии, подготовим датасеты и даталоадеры.

В классе `WordDataset` вам необходимо реализовать метод `__getitem__`, который будет возвращать сэмпл данных по входному idx, то есть список целых чисел (индексов слов).

Внутри этого метода необходимо добавить служебные токены начала и конца последовательности, а также токенизировать соответствующее предложение с помощью `word_tokenize` и сопоставить ему индексы из `word2ind`.

```
word2ind = {word: i for i, word in enumerate(vocab)}
ind2word = {i: word for word, i in word2ind.items()}

class WordDataset:
    def __init__(self, sentences):
        self.data = sentences
        self.unk_id = word2ind['<unk>']
        self.bos_id = word2ind['<bos>']
        self.eos_id = word2ind['<eos>']
```

```
        self.pad_id = word2ind['<pad>']

    def __getitem__(self, idx: int) -> List[int]:
        tokenized_sentence = []
        # Допишите код здесь
        tokenized_sentence += [self.bos_id]
        tokenized_sentence += [word2ind.get(word, self.unk_id) for
word in word_tokenize(self.data[idx])]
        tokenized_sentence += [self.eos_id]

        return tokenized_sentence

    def __len__(self) -> int:
        return len(self.data)

def collate_fn_with_padding(
    input_batch: List[List[int]], pad_id=word2ind['<pad>']) ->
torch.Tensor:
    seq_lens = [len(x) for x in input_batch]
    max_seq_len = max(seq_lens)

    new_batch = []
    for sequence in input_batch:
        for _ in range(max_seq_len - len(sequence)):
            sequence.append(pad_id)
        new_batch.append(sequence)

    sequences = torch.LongTensor(new_batch).to(device)

    new_batch = {
        'input_ids': sequences[:,:-1],
        'target_ids': sequences[:,1:]
    }

    return new_batch

train_sentences, eval_sentences = train_test_split(sentences,
test_size=0.2)
eval_sentences, test_sentences = train_test_split(sentences,
test_size=0.5)

train_dataset = WordDataset(train_sentences)
eval_dataset = WordDataset(eval_sentences)
test_dataset = WordDataset(test_sentences)

batch_size = 128

train_dataloader = DataLoader(
    train_dataset, collate_fn=collate_fn_with_padding,
batch_size=batch_size)
```

```
eval_dataloader = DataLoader(
    eval_dataset, collate_fn=collate_fn_with_padding,
batch_size=batch_size)

test_dataloader = DataLoader(
    test_dataset, collate_fn=collate_fn_with_padding,
batch_size=batch_size)
```

Обучение и архитектура модели

Вам необходимо на практике проверить, что влияет на качество языковых моделей. В этом задании нужно провести серию экспериментов с различными вариантами языковых моделей и сравнить различия в конечной перплексии на тестовом множестве.

Возможные идеи для экспериментов:

- Различные RNN-блоки, например, LSTM или GRU. Также можно добавить сразу несколько RNN блоков друг над другом с помощью аргумента num_layers. Вам поможет официальная документация здесь
- Различные размеры скрытого состояния. Различное количество линейных слоев после RNN-блока. Различные функции активации.
- Добавление нормализаций в виде Dropout, BatchNorm или LayerNorm
- Различные аргументы для оптимизации, например, подбор оптимального learning rate или тип алгоритма оптимизации SGD, Adam, RMSProp и другие
- Любые другие идеи и подходы

После проведения экспериментов необходимо составить таблицу результатов, в которой описан каждый эксперимент и посчитана перплексия на тестовом множестве.

Учтите, что эксперименты, которые различаются, например, только размером скрытого состояния или количеством линейных слоев считаются, как один эксперимент.

Успехов!

Функция evaluate (1 балл)

Заполните функцию `evaluate`

```
def evaluate(model, criterion, dataloader) -> float:

    model.eval()
    losses = []

    with torch.no_grad():
        for batch in dataloader:
            logits = model(batch['input_ids']).flatten(start_dim=0,
end_dim=1)
            loss = criterion(
                logits,
```

```
                batch['target_ids'].flatten()
            )
            losses.append(loss.item())

    mean_loss = sum(losses) / len(losses)
    perplexity = math.exp(mean_loss)

    return perplexity
```

Train loop (1 балл)

Напишите функцию для обучения модели.

```
import pandas as pd

df_stats = pd.DataFrame(columns=['Model', 'Loss', 'Perplexity',
'Diff_Loss', 'Diff_Perplexity', 'Better_than_base'])

display(df_stats)
```

```
Empty DataFrame
Columns: [Model, Loss, Perplexity, Diff_Loss, Diff_Perplexity,
Better_than_base]
Index: []
```

```
from IPython.display import clear_output

def train_model(model, criterion, optimizer,
                model_name='Current',
                plot=True,
                num_epoch=35,
                base_losses=None,
                base_perplexities=None,
                train_loader=train_dataloader,
                test_loader=eval_dataloader):

    # Напишите код здесь
    losses = []
    perplexities = []

    for epoch in range(num_epoch):
        epoch_losses = []
        model.train()

        for batch in tqdm(train_dataloader, desc=f'Training epoch
{epoch}'):
            optimizer.zero_grad()
            logits = model(batch['input_ids']).flatten(start_dim=0,
end_dim=1)
            loss = criterion(logits, batch['target_ids'].flatten())
```

```
            loss.backward()
            optimizer.step()

            epoch_losses.append(loss.item())

        losses.append(sum(epoch_losses) / len(epoch_losses))
        perplexities.append(evaluate(model, criterion,
eval_dataloader))

        # Отрисовка графиков
        if plot:

            fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(15,
6))

            if model_name != 'GRU_Base_Model':
                axes[0].plot(base_losses[:epoch+1], color='b',
marker='o', markersize=4, label='GRU_Base_Model')
            axes[0].plot(losses, color='r', marker='o', markersize=4,
label=model_name)

            axes[0].set_title('Losses')
            axes[0].set_xlabel('epoch')
            axes[0].set_ylabel('loss')
            axes[0].grid(True)
            axes[0].legend()

            if model_name != 'GRU_Base_Model':
                axes[1].plot(base_perplexities[:epoch+1], color='b',
marker='o', markersize=4, label='GRU_Base_Model')
            axes[1].plot(perplexities, color='r', marker='o',
markersize=4, label=model_name)

            axes[1].set_title('Perplexities')
            axes[1].set_xlabel('epoch')
            axes[1].set_ylabel('perplexity')
            axes[1].grid(True)
            axes[1].legend()

            clear_output(wait=True)
            plt.show()

        # Заполнение таблицы
        curr_model_loss = min(losses)
        curr_model_perp = min(perplexities)
        if model_name != 'GRU_Base_Model':
            diff_loss = curr_model_loss - min(base_losses)
            diff_perp = curr_model_perp - min(base_perplexities)
        else:
            diff_loss = 0
```

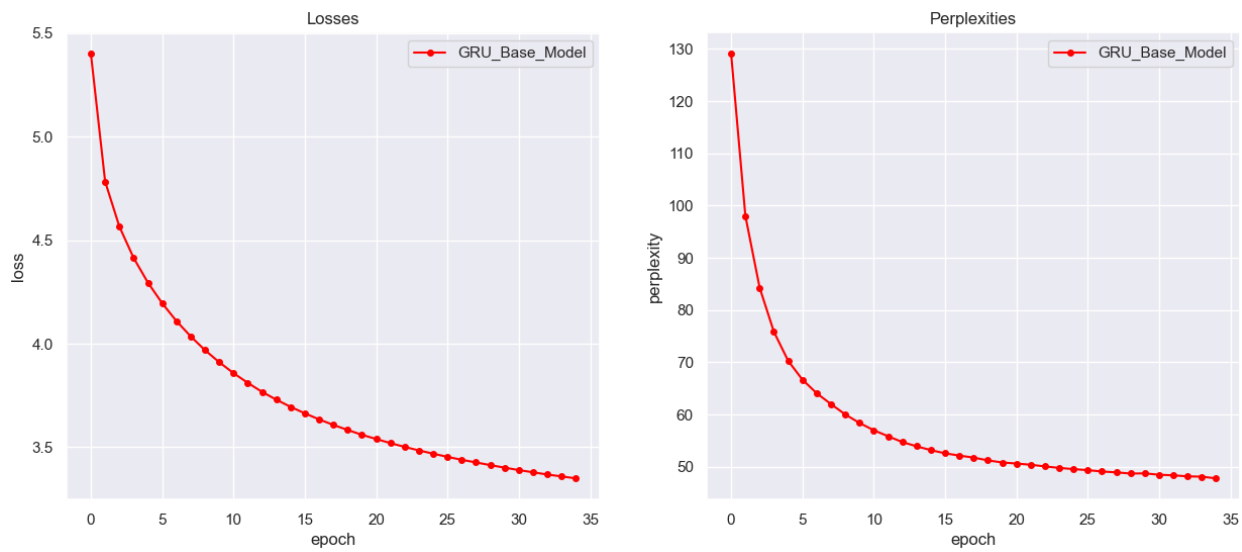
```
            diff_perp = 0

        if model_name != 'GRU_Base_Model':
            better_than_base = curr_model_perp <
min(base_perplexities)
        else:
            better_than_base = False

        data_to_add = (model_name, curr_model_loss, curr_model_perp,
diff_loss, diff_perp, better_than_base)
        if model_name in df_stats['Model'].values:
            index = df_stats[df_stats['Model'] == model_name].index
        else:
            index = len(df_stats)
        df_stats.loc[index] = data_to_add

    print(f'Loss = {curr_model_loss}')
    print(f'Perplexity = {curr_model_perp}')

    # Сравнение текущей модели с базовой моделью
    if model_name != 'GRU_Base_Model':
        print('\nСравнение с GRU_Base_Model:')
        print('Loss', ('Хуже', 'Лучше')[curr_model_loss <
min(base_losses)], 'на', abs(diff_loss))
        print('Perplexity', ('Хуже', 'Лучше')[curr_model_perp <
min(base_perplexities)], 'на', abs(diff_perp))

    return losses, perplexities
```

Base model

В качестве базовой модели возьмем модель из семинара. Результаты моделей из дальнейших экспериментов будем сравнивать с базовой моделью. Оценка того, какая из моделей лучше, будет основываться на метрике Perplexity.

```
class GRU_Base_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)  # [batch_size,
seq_len, hidden_dim]
```

```
        output, _ = self.rnn(embeddings)  # [batch_size, seq_len,
hidden_dim]
        output = self.dropout(self.linear(self.non_lin(output)))  #
[batch_size, seq_len, hidden_dim]
        projection = self.projection(self.non_lin(output))  #
[batch_size, seq_len, vocab_size]

        return projection

model_name='GRU_Base_Model'

model = GRU_Base_LM(hidden_dim=256, vocab_size=len(vocab)).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

base_losses, base_perplexities = train_model(model=model,
                                             criterion=criterion,
                                             optimizer=optimizer,
                                             model_name=model_name)
```




```
Loss = 3.3486153530863536
Perplexity = 47.7399529976823

display(df_stats)
```

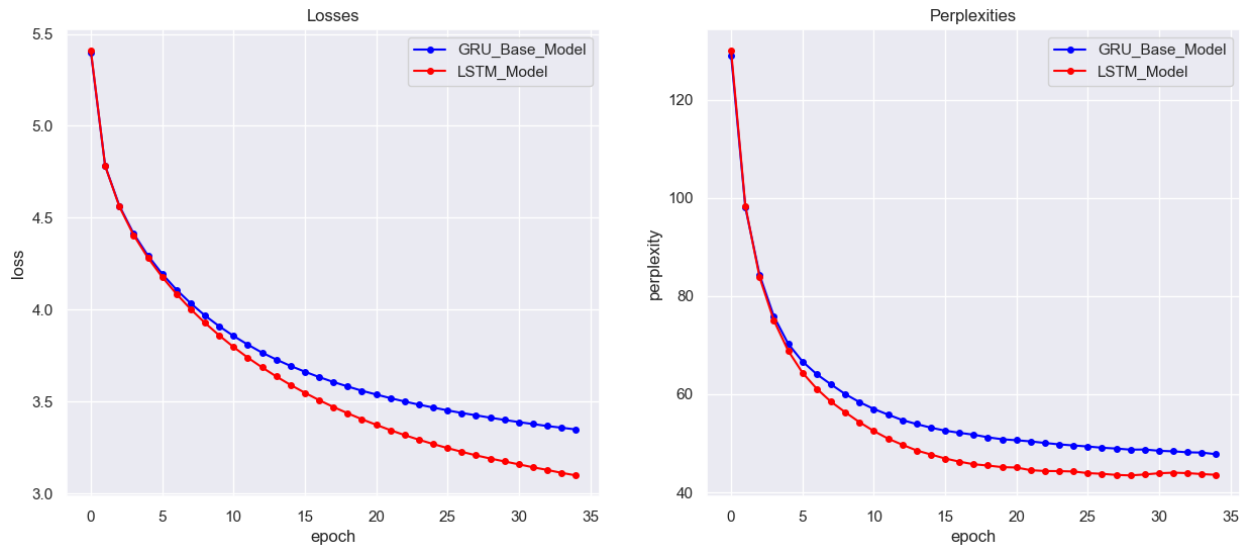
```
            Model      Loss  Perplexity  Diff_Loss  Diff_Perplexity  \
0  GRU_Base_Model  3.348615   47.739953          0                0

   Better_than_base
0             False
```

Замена GRU на LSTM (Первый эксперимент (2 балла))

Определите архитектуру модели и обучите её.

LSTM в отличии GRU лучше запоминает длинные зависимости.

```
class LSTM_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int, num_layers:
int = 1):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.LSTM(hidden_dim, hidden_dim,
num_layers=num_layers, batch_first=True)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'LSTM_Model'

model = LSTM_LM(hidden_dim=256, vocab_size=len(vocab),
num_layers=1).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

lstm_losses, lstm_perplexities = train_model(model=model,
                                             criterion=criterion,
                                             optimizer=optimizer,
                                             model_name=model_name,
                                             plot=True,
                                             num_epoch=35,
                                             base_losses=base_losses,

base_perplexities=base_perplexities)
```




```
Loss = 3.0989123436106087
Perplexity = 43.42439667733452

Сравнение с GRU_Base_Model:
Loss Лучше на 0.24970300947574486
Perplexity Лучше на 4.315556320347781

display(df_stats)
```

```
            Model      Loss  Perplexity  Diff_Loss  Diff_Perplexity  \
0  GRU_Base_Model  3.348615   47.739953   0.000000         0.000000
1      LSTM_Model  3.098912   43.424397  -0.249703        -4.315556

   Better_than_base
0             False
1              True
```

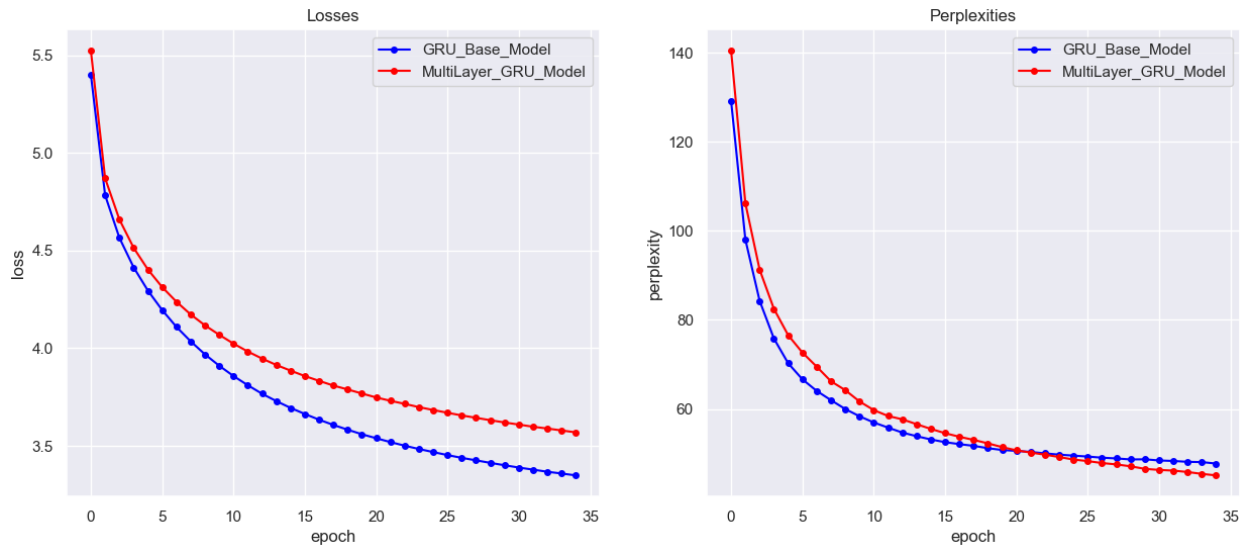
Использование сразу нескольких RNN блоков (Второй эксперимент (2 балла))

Попробуйте что-то поменять в модели или в пайплайне обучения, идеи для экспериментов можно подсмотреть выше.

Два блока RNN num_layers = 2. Позволит выделить более сложные иерархические зависимости.

```
class MultiLayerGRU_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int, num_layers:
int = 2):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim,
```

```
num_layers=num_layers, batch_first=True, dropout=0.1)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'MultiLayer_GRU_Model'

model = MultiLayerGRU_LM(hidden_dim=256, vocab_size=len(vocab),
num_layers=2).to(device) # num_layers=2
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

multilayer_losses, multilayer_perplexities = train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,
                                                                plot=True,
                                                                num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```




```
Loss = 3.5687269679212994
Perplexity = 45.088227082194905

Сравнение с GRU_Base_Model:
Loss Хуже на 0.2201116148349458
Perplexity Лучше на 2.651725915487397

display(df_stats)

                  Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0        GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1            LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2  MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726

   Better_than_base
0             False
1              True
2              True
```

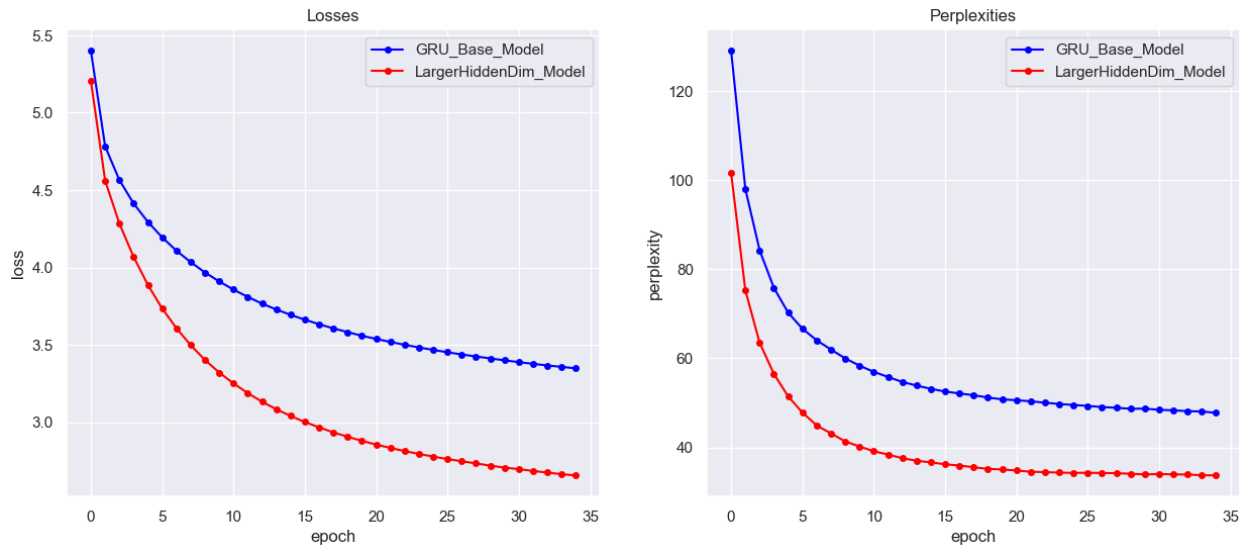
Далее будут ещё эксперименты

Увеличенние размера скрытого состояния

hidden_dim=512 позволит изучить более сложные закономерности.

```
class LargerHiddenDim_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
```

```
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'LargerHiddenDim_Model'

model = LargerHiddenDim_LM(hidden_dim=512,
vocab_size=len(vocab)).to(device) # hidden_dim=512
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

larger_hidden_losses, larger_hidden_perplexities =
train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,

plot=True,

num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```

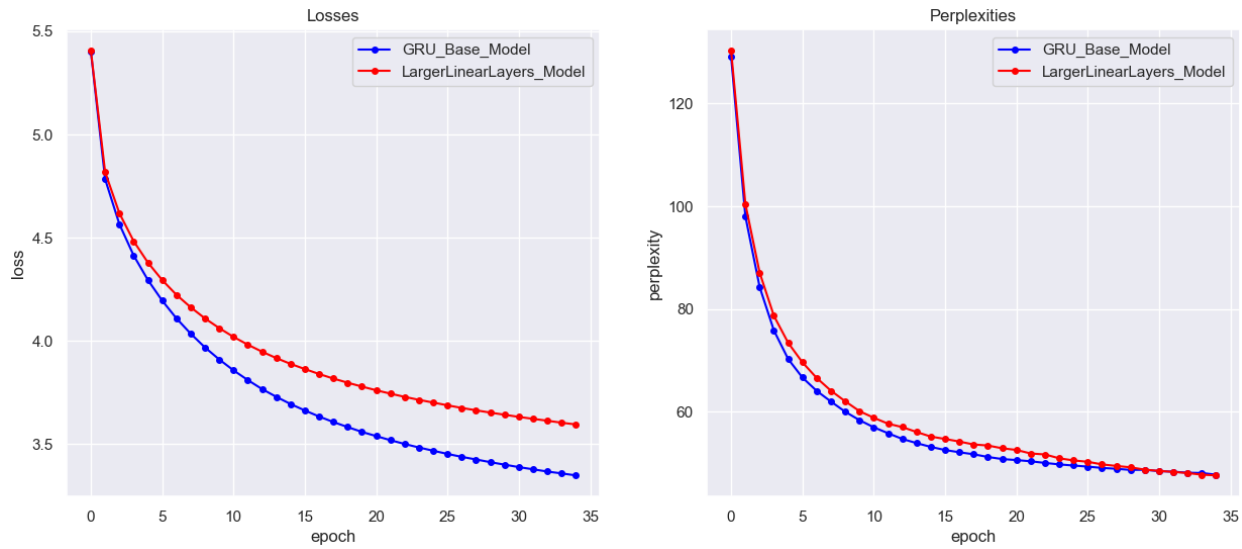



```
Loss = 2.65725870489211
Perplexity = 33.709729575689025

Сравнение с GRU_Base_Model:
Loss Лучше на 0.6913566481942435
Perplexity Лучше на 14.030223421993277

display(df_stats)

                    Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0          GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1              LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2   MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3  LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223

   Better_than_base
0             False
1              True
2              True
3              True
```

Увеличенние количества линейных слоёв после RNN

Модель с двумя линейными слоями. Теоретически позволит лучше распредлять вероятность следования слов.

```python
class LargerLinearLayers_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)

        # Два линейных слоя
        self.linear_layers = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim),
            nn.Tanh(),
            nn.Dropout(p=0.1),
            nn.Linear(hidden_dim, hidden_dim),
            nn.Tanh(),
            nn.Dropout(p=0.1)
        )
        self.projection = nn.Linear(hidden_dim, vocab_size)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.linear_layers(output)
        projection = self.projection(output)

        return projection

model_name = 'LargerLinearLayers_Model'

model = LargerLinearLayers_LM(hidden_dim=256,
vocab_size=len(vocab)).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

larger_linear_losses, larger_linear_perplexities =
train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,

plot=True,

num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```

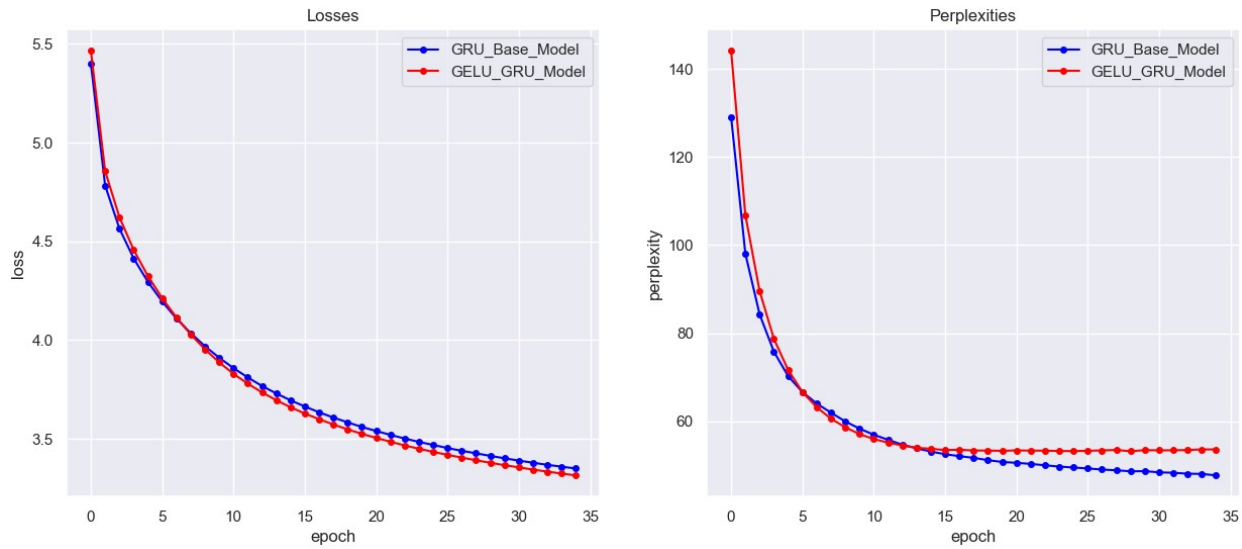



```
Loss = 3.595223607886538
Perplexity = 47.607385705066925

Сравнение с GRU_Base_Model:
Loss Хуже на 0.24660825480018422
Perplexity Лучше на 0.13256729261537714

display(df_stats)

                       Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0             GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608        -
0.132567

   Better_than_base
0             False
1              True
2              True
3              True
4              True
```

Использование альтернативной функции активации (GELU)

Применим функцию активации GELU вместо Tanh. Gaussian Error Linear Unit учитывает вероятностный характер нейронов.

```python
class GELU_GRU_LM(nn.Module):

    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.GELU()    # GELU вместо Tanh
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'GELU_GRU_Model'

model = GELU_GRU_LM(hidden_dim=256, vocab_size=len(vocab)).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

gelu_losses, gelu_perplexities = train_model(model=model,
                                             criterion=criterion,
                                             optimizer=optimizer,
                                             model_name=model_name,
                                             plot=True,
                                             num_epoch=35,
                                             base_losses=base_losses,

base_perplexities=base_perplexities)
```

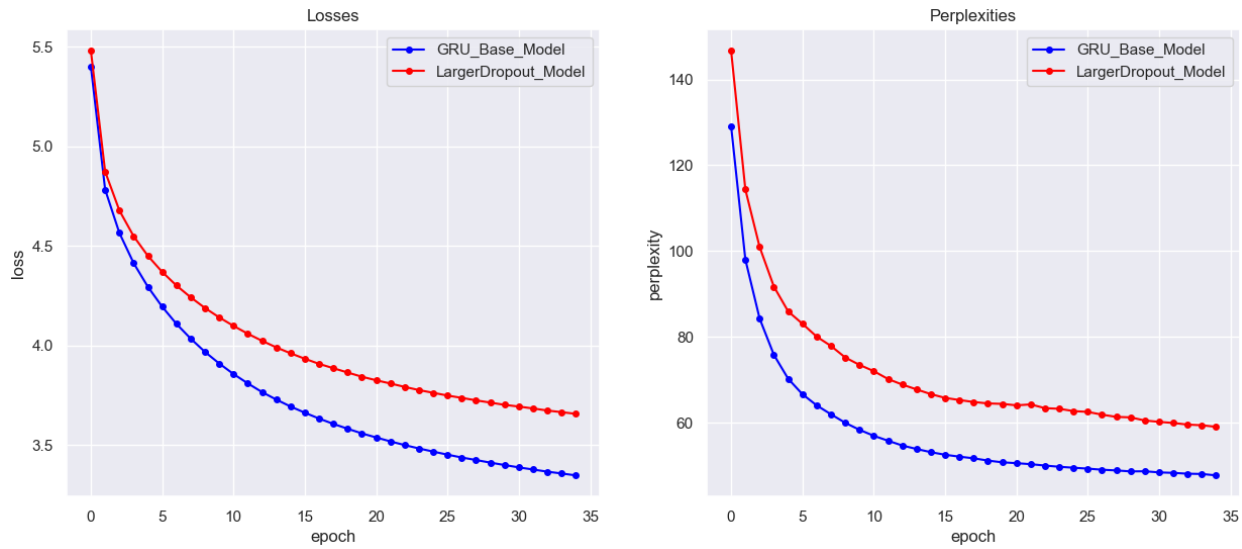



```
Loss = 3.3139008934806644
Perplexity = 53.21709448245683

Сравнение с GRU_Base_Model:
Loss Лучше на 0.034714459605689196
Perplexity Хуже на 5.477141484774528

display(df_stats)

                      Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0            GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608        -
0.132567
5            GELU_GRU_Model  3.313901   53.217094  -0.034714
5.477141

   Better_than_base
0             False
1              True
2              True
3              True
4              True
5             False
```

Применение различных нормализаций

Увеличенние Dropout для регуляризации. Попробуем dropout = 0.3, тем самым уменьшим вероятность переобучения.

```
class LargerDropout_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int, dropout_rate:
float = 0.3):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True,
dropout=dropout_rate)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=dropout_rate)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'LargerDropout_Model'

model = LargerDropout_LM(hidden_dim=256, vocab_size=len(vocab),
dropout_rate=0.3).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

larger_dropout_losses, larger_dropout_perplexities =
train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,

plot=True,

num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```

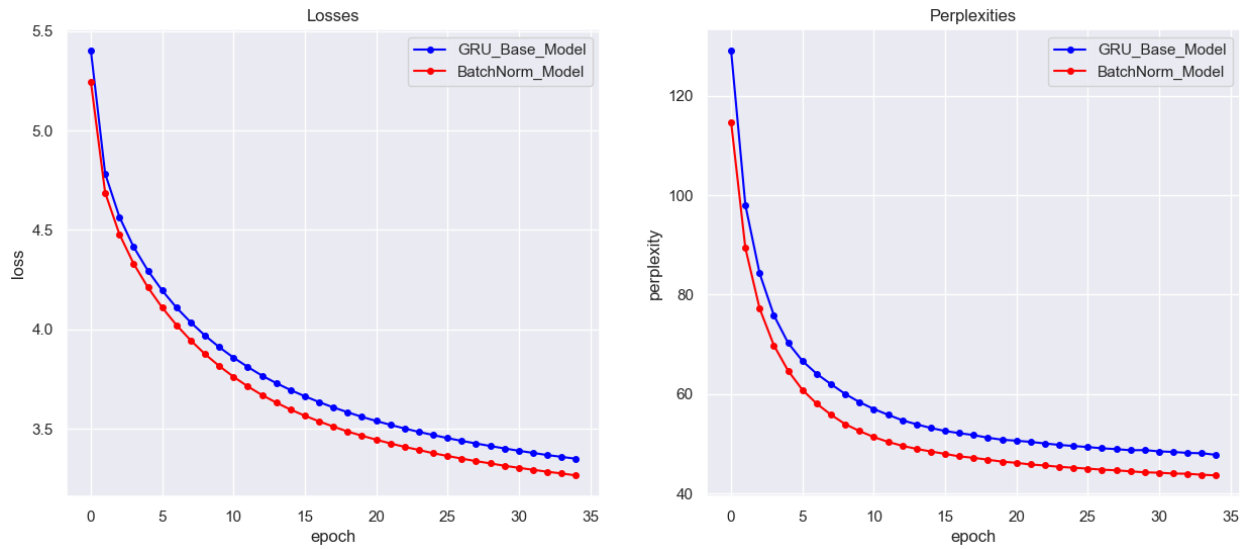



```
Loss = 3.6574581943014794
Perplexity = 59.0532194927761

Сравнение с GRU_Base_Model:
Loss Хуже на 0.30884284121512584
Perplexity Хуже на 11.3132664950938

display(df_stats)

                       Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0             GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608        -
0.132567
5             GELU_GRU_Model  3.313901   53.217094  -0.034714
5.477141
6       LargerDropout_Model  3.657458   59.053219   0.308843
11.313266

   Better_than_base
0             False
1              True
2              True
3              True
4              True
```

```
5              False
6              False
```

Добавление BatchNorm1d после RNN.

```
class BatchNorm_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)
        self.batch_norm = nn.BatchNorm1d(hidden_dim)    # BatchNorm
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = output.transpose(1, 2)
        output = self.batch_norm(output)
        output = output.transpose(1, 2)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'BatchNorm_Model'

model = BatchNorm_LM(hidden_dim=256, vocab_size=len(vocab)).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

batchnorm_losses, batchnorm_perplexities = train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,
                                                        plot=True,
                                                        num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```

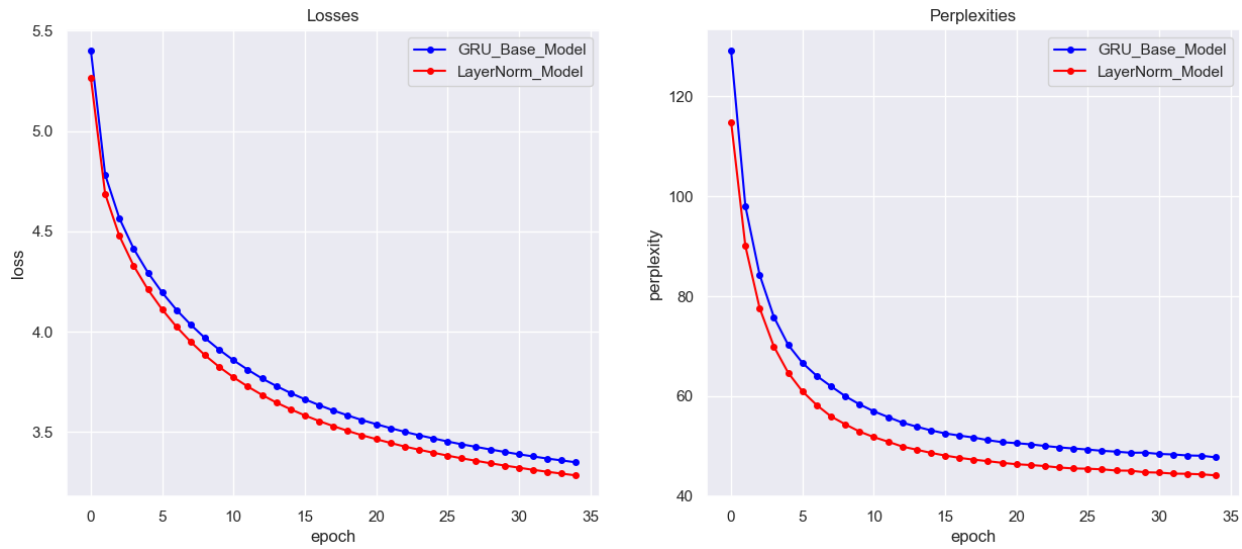



```
Loss = 3.2650455097373414
Perplexity = 43.608647857048105

Сравнение с GRU_Base_Model:
Loss Лучше на 0.08356984334901219
Perplexity Лучше на 4.131305140634197

display(df_stats)

                      Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0            GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608        -
0.132567
5            GELU_GRU_Model  3.313901   53.217094  -0.034714
5.477141
6       LargerDropout_Model  3.657458   59.053219   0.308843
11.313266
7           BatchNorm_Model  3.265046   43.608648  -0.083570        -
4.131305

   Better_than_base
0             False
1              True
2              True
3              True
```

```
4                True
5               False
6               False
7                True
```

Добавление LayerNorm после RNN. Предпочтительнее для моделей RNN, нежели BatchNorm.

```
class LayerNorm_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.GRU(hidden_dim, hidden_dim, batch_first=True)
        self.layer_norm = nn.LayerNorm(hidden_dim)  # LayerNorm
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.layer_norm(output)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'LayerNorm_Model'

model = LayerNorm_LM(hidden_dim=256, vocab_size=len(vocab)).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

layernorm_losses, layernorm_perplexities = train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,
                                                                plot=True,
                                                                num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```




```
Loss = 3.2834822053594737
Perplexity = 44.143426833442824

Сравнение с GRU_Base_Model:
Loss Лучше на 0.06513314772687995
Perplexity Лучше на 3.596526164239478

display(df_stats)

                       Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0             GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703          -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112          -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357         -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608          -
0.132567
5            GELU_GRU_Model  3.313901   53.217094  -0.034714
5.477141
6       LargerDropout_Model  3.657458   59.053219   0.308843
11.313266
7           BatchNorm_Model  3.265046   43.608648  -0.083570          -
4.131305
8           LayerNorm_Model  3.283482   44.143427  -0.065133          -
3.596526

   Better_than_base
0             False
1              True
```

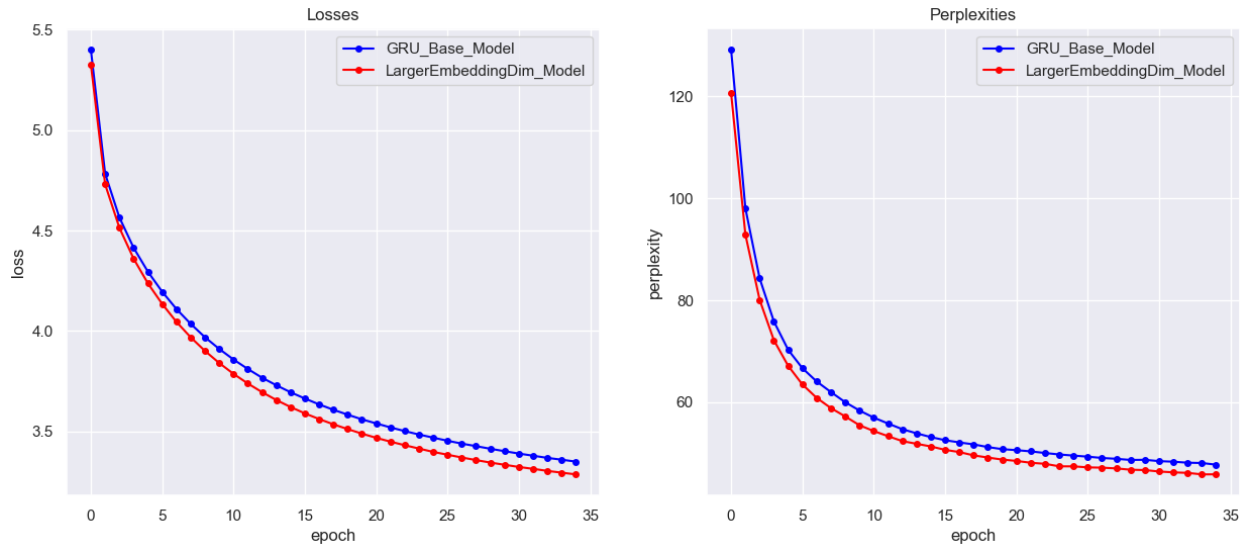
```
2                True
3                True
4                True
5               False
6               False
7                True
8                True
```

Увеличенние размера эмбеддингов

embedding_dim=512 для увеличения векотора предствления каждого слова. Для каждого слова кодируется больше информации.

```
class LargerEmbeddingDim_LM(nn.Module):
    def __init__(self, embedding_dim: int, hidden_dim: int,
vocab_size: int):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embedding_dim)
        self.rnn = nn.GRU(embedding_dim, hidden_dim, batch_first=True)
        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=0.1)

    def forward(self, input_batch) -> torch.Tensor:
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)
        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'LargerEmbeddingDim_Model'

model = LargerEmbeddingDim_LM(embedding_dim=512, hidden_dim=256,
vocab_size=len(vocab)).to(device) # embedding_dim=512
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())

larger_emb_losses, larger_emb_perplexities = train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,
                                                                plot=True,

num_epoch=35,
```

```
base_losses=base_losses,

base_perplexities=base_perplexities)
```




```
Loss = 3.285025845586726
Perplexity = 45.840118876754204

Сравнение с GRU_Base_Model:
Loss Лучше на 0.06358950749962755
Perplexity Лучше на 1.899834120928098

display(df_stats)

                       Model      Loss  Perplexity  Diff_Loss
Diff_Perplexity  \
0             GRU_Base_Model  3.348615   47.739953   0.000000
0.000000
1                LSTM_Model  3.098912   43.424397  -0.249703        -
4.315556
2      MultiLayer_GRU_Model  3.568727   45.088227   0.220112        -
2.651726
3     LargerHiddenDim_Model  2.657259   33.709730  -0.691357       -
14.030223
4  LargerLinearLayers_Model  3.595224   47.607386   0.246608        -
0.132567
5             GELU_GRU_Model  3.313901   53.217094  -0.034714
5.477141
6       LargerDropout_Model  3.657458   59.053219   0.308843
11.313266
7           BatchNorm_Model  3.265046   43.608648  -0.083570        -
4.131305
```

```
8           LayerNorm_Model  3.283482   44.143427  -0.065133        -
3.596526
9  LargerEmbeddingDim_Model  3.285026   45.840119  -0.063590        -
1.899834

   Better_than_base
0             False
1              True
2              True
3              True
4              True
5             False
6             False
7              True
8              True
9              True
```

Финальная модель (комбинирование подходов)

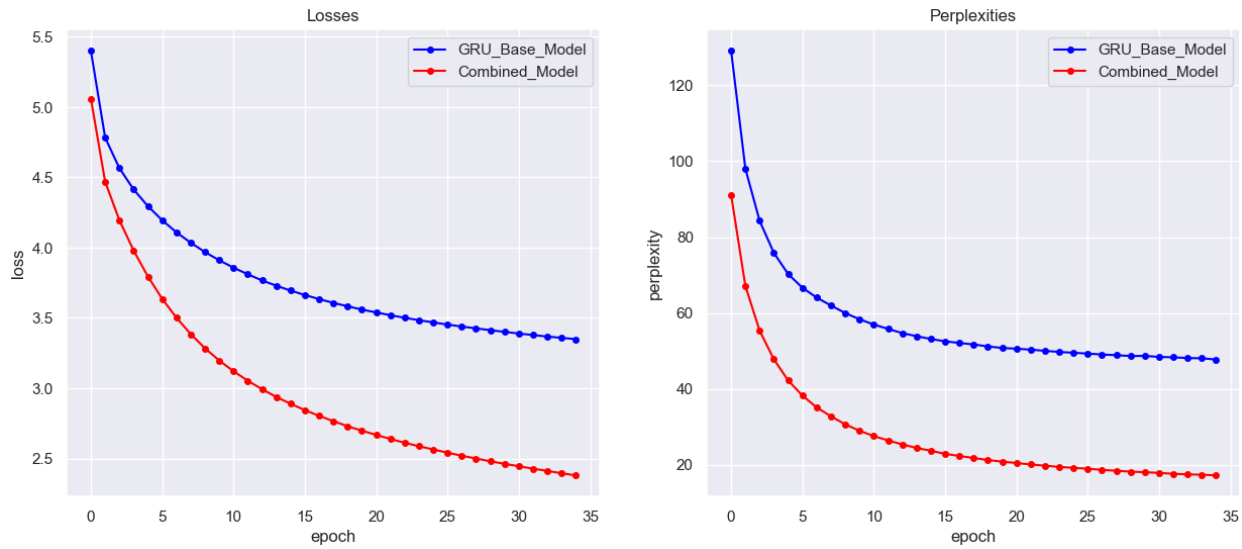
Хорошо показали себя подходы:

- Замена GRU на LSTM
- Использование сразу нескольких RNN блоков
- Увеличение размера скрытого состояния
- Применения нормализаций BatchNom и LayerNorm (выберем LayerNorm, потому что данная нормализация является лучшей практикой для RNN моделей)

Применим подходы в финальной модели:

```
class Combined_LM(nn.Module):
    def __init__(self, hidden_dim: int, vocab_size: int, num_layers:
int = 1, dropout_rate: float = 0.1):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, hidden_dim)
        self.rnn = nn.LSTM(hidden_dim, hidden_dim,
num_layers=num_layers,
                           batch_first=True, dropout=dropout_rate if
num_layers > 1 else 0)

        # self.batch_norm = nn.BatchNorm1d(hidden_dim)    # BatchNorm
        self.layer_norm = nn.LayerNorm(hidden_dim)      # LayerNorm

        self.linear = nn.Linear(hidden_dim, hidden_dim)
        self.projection = nn.Linear(hidden_dim, vocab_size)

        self.non_lin = nn.Tanh()
        self.dropout = nn.Dropout(p=dropout_rate)

    def forward(self, input_batch) -> torch.Tensor:
```

```
        embeddings = self.embedding(input_batch)
        output, _ = self.rnn(embeddings)

        # output = output.transpose(1, 2)    # BatchNorm
        # output = self.batch_norm(output)    # BatchNorm
        # output = output.transpose(1, 2)    # BatchNorm

        output = self.layer_norm(output)    # LayerNorm

        output = self.dropout(self.linear(self.non_lin(output)))
        projection = self.projection(self.non_lin(output))

        return projection

model_name = 'Combined_Model'

model = Combined_LM(hidden_dim=512, vocab_size=len(vocab),
num_layers=2).to(device)
criterion = nn.CrossEntropyLoss(ignore_index=word2ind['<pad>'])
optimizer = torch.optim.Adam(model.parameters())
# optimizer = torch.optim.AdamW(model.parameters(), lr=0.001,
weight_decay=0.01)

combined_losses, combined_perplexities = train_model(model=model,

criterion=criterion,

optimizer=optimizer,

model_name=model_name,
                                                        plot=True,
                                                        num_epoch=35,

base_losses=base_losses,

base_perplexities=base_perplexities)
```




```
Loss = 2.379735365378118
Perplexity = 17.22332997258968

Сравнение с GRU_Base_Model:
Loss Лучше на 0.9688799877082355
Perplexity Лучше на 30.51662302509262

display(df_stats)
```

	Model	Loss	Perplexity	Diff_Loss	Diff_Perplexity	Better_than_base
0	GRU_Base_Model	3.348615	47.739953	0.000000	0.000000	False
1	LSTM_Model	3.098912	43.424397	-0.249703	-4.315556	True
2	MultiLayer_GRU_Model	3.568727	45.088227	0.220112	-2.651726	True
3	LargerHiddenDim_Model	2.657259	33.709730	-0.691357	-14.030223	True
4	LargerLinearLayers_Model	3.595224	47.607386	0.246608	-0.132567	True
5	GELU_GRU_Model	3.313901	53.217094	-0.034714	5.477141	False
6	LargerDropout_Model	3.657458	59.053219	0.308843	11.313266	False
7	BatchNorm_Model	3.265046	43.608648	-0.083570	-4.131305	True
8	LayerNorm_Model	3.283482	44.143427	-0.065133	-3.596526	True
9	LargerEmbeddingDim_Model	3.285026	45.840119	-0.063590		
10	Combined_Model	2.379735	17.223330	-0.968880		

```
9          -1.899834              True
10        -30.516623              True
```

Отчет (2 балла)

Опишите проведенные эксперименты. Сравните перплексии полученных моделей. Предложите идеи по улучшению качества моделей.

Было реализовано:

- Базовая модель с одним GRU слоем. Все последующие реализации сравнивались с метриками данныой модели.
- Была опробована замена GRU слоя на LSTM слой. Модель показала значительно лучшие результаты. Модель борется с затухающими градиентами и лучше запоминает длинные зависимости.
- Использования двух GRU слоев. Заметно улучшение метрики Perplexity. Результат, возможно, будет лучше в связке с регуляризацями.
- Увеличение скрытого состояния с 256 до 512. Лучший результат. Модель смогла выучить более сложные зависимости.
- Добавление дополнительного линейного слоя и небольшим Dropout. Значение метрик хуже, чем в базовой модели.
- Замена функции активации Tanh на GELU. Значение метрик хуже, чем в базовой модели.
- Увеличение Dropout с 0.1 до 0.3. Значение метрик хуже, чем в базовой модели. Возможно, что модель недообучилась и необходим подбор оптимального значения Dropout.
- Добавлена Batch Normalization (нормализация активации по батчу). Значение метрик лучше, чем в базовой модели.
- Добавлена Layer Normalization (нормализация активации по каждому элементу батча). Значение метрик лучше, чем в базовой модели. Для RNN моделей предпочтительнее.
- Увеличение размерности эмбеддингов с 256 до 512. Значимого улучшения не последовало.

Реализация финальной модели Combined_Model, которая включает лучшие практики из числа проведенных экспериментов:

- Замена GRU слоя на LSTM слой
- Использования двух слоев RNN (два слоя LSTM)
- Использование размера скрытого слоя hidden_dim = 512
- Добавлена Layer Normalization

Модель показала лучшие значения метрики Perplexity (17.223330); Loss (2.379735) также оказался самым низким относительно Loss'а других моделей.

Предложения по улучшению:

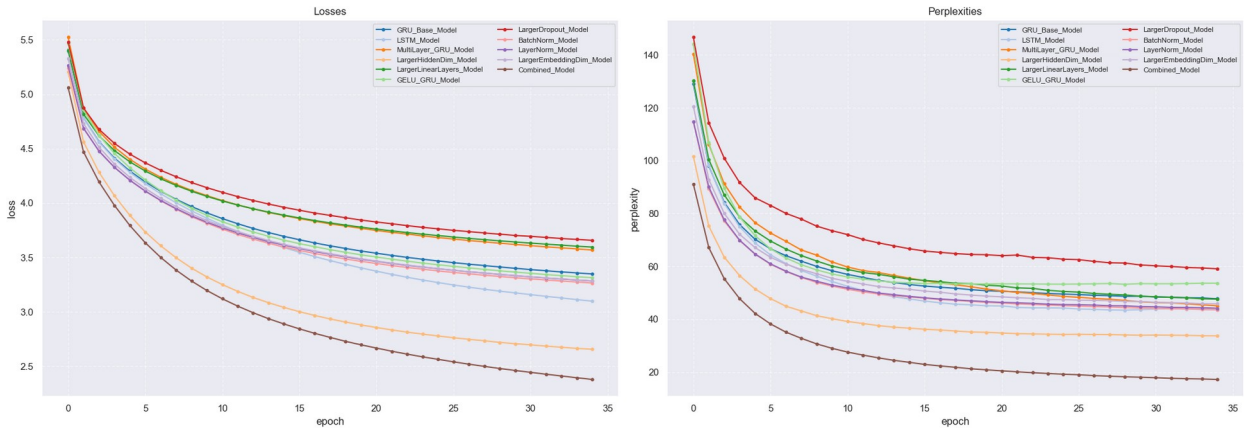
- Подобрать оптимальный оптимизатор
- Подобрать оптимальные гиперпараметры

- Реализовать Early Stopping

График сравнения всех моделей

```
fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(20, 7))

# Словарь со всеми моделями и полученными метриками
histories = {
    'GRU_Base_Model': (base_losses, base_perplexities),
    'LSTM_Model': (lstm_losses, lstm_perplexities),
    'MultiLayer_GRU_Model': (multilayer_losses,
multilayer_perplexities),
    'LargerHiddenDim_Model': (larger_hidden_losses,
larger_hidden_perplexities),
    'LargerLinearLayers_Model': (larger_linear_losses,
larger_linear_perplexities),
    'GELU_GRU_Model': (gelu_losses, gelu_perplexities),
    'LargerDropout_Model': (larger_dropout_losses,
larger_dropout_perplexities),
    'BatchNorm_Model': (batchnorm_losses, batchnorm_perplexities),
    'LayerNorm_Model': (layernorm_losses, layernorm_perplexities),
    'LargerEmbeddingDim_Model': (larger_emb_losses,
larger_emb_perplexities),
    'Combined_Model': (combined_losses, combined_perplexities),
}

colors = plt.cm.tab20.colors

for i, (model_name, (losses, perplexities)) in
enumerate(histories.items()):
    color = colors[i % len(colors)]

    axes[0].plot(losses, color=color, marker='o', markersize=3,
linewidth=1.5, label=model_name)
    axes[1].plot(perplexities, color=color, marker='o', markersize=3,
linewidth=1.5, label=model_name)

axes[0].set_title('Losses')
axes[0].set_xlabel('epoch')
axes[0].set_ylabel('loss')
axes[0].grid(True, linestyle='--', alpha=0.6)
axes[0].legend(loc='upper right', fontsize='x-small', ncol=2)

axes[1].set_title('Perplexities')
axes[1].set_xlabel('epoch')
axes[1].set_ylabel('perplexity')
axes[1].grid(True, linestyle='--', alpha=0.6)
axes[1].legend(loc='upper right', fontsize='x-small', ncol=2)
```

```
plt.tight_layout()

plt.show();
```



Проверка на тесте

```
test_perplexity = evaluate(model, criterion, test_dataloader)

print(f'Perplexity на тесте: {test_perplexity}')

Perplexity на тесте: 17.3304825229429
```

Попробуем сгенерировать продолжение предложений

```
def generate_sequence(model, starting_seq: str, max_seq_len: int = 50)
-> str:
    device = next(model.parameters()).device
    model.eval()

    # Разбиваем начальную строку на слова
    starting_words = starting_seq.strip().split()

    input_ids = [word2ind['<bos>']] + [
        word2ind.get(word, word2ind['<unk>']) for word in
starting_words]
    input_ids = torch.LongTensor(input_ids).to(device)

    with torch.no_grad():
        for i in range(max_seq_len):
            output = model(input_ids.unsqueeze(0))
            next_word_distribution = output[0, -1, :]
            next_word = next_word_distribution.argmax()
            input_ids = torch.cat([input_ids, next_word.unsqueeze(0)])

            if next_word.item() == word2ind['<eos>']:
                break
```

```
    words = ' '.join([ind2word[idx.item()] for idx in input_ids[1:]
                      if ind2word[idx.item()] not in ['<pad>',
'<unk>', '<eos>']])

    return words

generate_sequence(model, starting_seq='The filmmakers tried hard, but
')
```

```
"filmmakers tried but were n't much better ."
```

```
generate_sequence(model, starting_seq='never seen so many mistakes')
```

```
'never seen so many mistakes in this film .'
```

```
generate_sequence(model, starting_seq='this film')
```

```
'this film is a great movie .'
```